nature portfolio

Peer Review File

A longitudinal cohort study uncovers plasma protein biomarkers predating clinical onset and treatment response of rheumatoid arthritis

Corresponding Author: Dr Lunzhi Dai

This file contains all reviewer reports in order by version, followed by all author rebuttals in order by version.

Version 0:

Reviewer comments:

Reviewer #1

(Remarks to the Author)
A longitudinal cohort study uncovers plasma protein biomarkers predating clinical onset and treatment response of rheumatoid arthritis

This is an interesting proteomics study from He et al. which uses a well-described cohort of RA patients, at-risk individuals, and healthy controls, to better understand plasma protein changes associated with the various states. They main results suggests that there are overlapping differences in the plasma proteome between RA and at-risk individuals, proteins that correlate with disease activity, and proteins that predict treatment response with conventional DMARDs, some of which change in individuals who respond to the drug in follow up testing.

The strength of the study includes:

1. A well characterized, large cohort of individuals are included, along with pre-post samples after treatment with csDMARDs
2. An exhaustive bioinformatics analysis to address several clinically important and exploratory questions
3. Tandem mass tagging to reduce batch effect and more accurately quantify protein levels. The methods appear to be sound.
4. Inclusion of treatment naive RA patients
5. Sex-considerations are made in key aspects of the manuscript

The main drawback of the paper is that no validation methods are deployed to ensure that accessible methodology (ELISA, or other antibody-based methods) is in agreement with these results. There are no mechanistic studies and no external validation cohort, which limits the generalizability of the results. The methods are well described, the work is original and could be a significant resource for follow up studies.

Major
-the title speaks to a longitudinal study, with reference to pre-RA. While it’s extremely valuable to consider the pre-RA state, these samples are analyzed in cross-section, not longitudinally (ie. the individuals have not progressed into inflammatory arthritis). A change in the title is required.
-how many of the ACPA+ at-risk have developed RA? An analysis that aims to distinguish those who progress vs those who have not progressed would be valuable, if the numbers were high enough.
-more details pertaining to the ACPA+ at-risk are required. Are there clinical data such as joint pain and stiffness? VAS? What are the ACPA levels of these individuals?
-Figure 1 lays out the experimental design well but all of the quality control figures should be considered for supplemental data
-most TLRs are either expressed intracellularly on the plasma membrane ,yet these are upregulated in individuals with RA - where are they coming from?
-the ACPA+ RA patients had higher disease activity than the ACPA- RA patients - is this the primary driver of the inflammatory response signatures or was disease activity controlled for?

-the analysis on disease activity trends with age is problematic. 1. CRP tends to increase with age, which is an interesting phenomenon but may increase DAS28-CRP levels for different reasons outside of RA activity. 2. Tender joint counts may be a reflection of joint damage, rather than active joint swelling. I agree however with controlling for age and sex, as was done in the next analysis.
-the non-linear analysis is of interest, but the 2 methods employed seem overlapping and somewhat redundant. The agreement between DE-SWAN and the clinically important cut offs for DAS28 are reassuring, however this seems like an overly exhaustive analysis without a specific direction.
-it's not clear to me why MTX-LEF combo therapy is used in such high rates, is this combination starting together at the first visit, or is LEF being added to MTX if MTX monotherapy is not successful. It is widely known that MTX-LEF is a challenging combination of treatments to use in RA due to their effects on the liver.
-The results on predicting treatment responses are appealing, due to the high ROC and low # of inputs required. Can these proteins be measured using multiple reaction monitoring or ELISA as a follow up?

Minor
-line 52: "at-risk" includes other individuals, namely first-degree relatives, individuals carrying shared epitope, etc.
-line 116: does this reference show that neuronal expansion pathways are upregulated in at-risk individuals?
-line 131: gender or sex?

Reviewer #2

(Remarks to the Author)
• The manuscript by He et al, investigates the potential of identifying plasma biomarkers of Disease response and possibly pre-disease onset. Predicting disease onset and response is one of the key unmet needs in the field of Rheumatology.

• However, I do have a few queries.

• There is not enough information on how the authors defined individuals at risk. They state ACPA+ and aches/pains with no clinical sign of disease activity. However, they need to be more precise in their exact description, IAR should be categorised according to EULAR criteria where clinical assessment of these individuals gives a score <6, thus not reaching criteria for definitive RA. They should include any clinical details that were assessed, is CRP, SJC, TJC, VAS. Also did they use ultrasound guided analysis to define definitively that there was no joint inflammation. One or both of these approaches are the goal standard for defining IAR.

• The authors do discuss effect of age and gender. The IAR group is much younger than the RA and HC groups, did they take this into account. They should do a sub-analysis comparing aged-matched HC, RA and IAR groups.

• The authors also discuss differences in gender in the RA group. The % females in the IAR group are different from RA group (60% vs 85%) therefore they should perform a sub-analysis.

• For Figure 2, they show volcano blots comparing two groups at a time, it would be more useful to perform unsupervised hierarchical clustering for all 4 groups to see if they separate out into distinct phenotype groups based on plasma proteins or is there overlap.

• Figure 2d, are the authors surprised there is a divergence in the level of some proteins between IAR and RA, would they not expect that as IAR is pre-disease onset they might just all follow stepwise progression from HC.

• In Figure 2d, they show HC vs IAR vs all RA, then in Figure 2f and g they compare HC vs RA ACPA-neg vs RA ACPA-pos. Considering IAR are also positive, inclusion of this cohort is important for this figure to investigate if IAR differ against ACPA-neg and ACPA-pos.

• Figure 3- focuses on gender, and observe differences, which is expected but nice to see. Again, here based on the differences in % gender in IAR group, analysis of this group would be important to examine in this context.

• Did they expect the DAs28 to be higher in males, it tends to be more associated with females? Were correlations maintained if they split the group into ACPA Pos and ACPA-neg and separately analysed.

• In figure 4 they show 6 specific gene clusters. Then perform unsupervised hierarchical clustering which didn't actually show specific separation. When they perform a wave DAS 28 clustering the authors suggest there are three peaks 2.9, 4.0 and 5.1. However, the clustering in the heatmap does not seem that definitive. What do they think the significance of these three peaks are regarding clinical practice and assessment.

• Figure 5-6, identify proteins associated with response. Firstly, it would be interesting to analyse ACPA pos and ACPA neg separately, as it is thought they respond differently. Secondly based on their gender data, analysis of female vs males' responses would also be important, as it is known males respond better.

• Finally, where the paper is lacking is the further analysis of the IAR group throughout. In particular, how many of the IAR patients converted to RA vs non-convertors. This really is the key question. Had convertors higher levels of protein, did those that converted earlier have higher levels. Do the authors have plasma from those when they converted vs those that did not.

• The authors are suggesting the markers identified are potential protein biomarkers for RA. Do they think they are specific to RA, i.e., are these markers generally up in inflammatory diseases, i.e., other rheumatic diseases like PsA, Ank Spon, outside the field like inflammatory bowel disease, psoriasis. Think this would be useful.

• Would they use one protein or a panel of proteins?

• They would need to validate the panel in an additional cohort.

Version 1:

Reviewer comments:

Reviewer #1

(Remarks to the Author)
The authors have undertaken a significant revision of their manuscript and most of my critiques have been addressed.

I only have one minor comment:
-if the ACPA+ at-risk individuals were not assessed for tender joints or pain VAS, it is not appropriate to score these as 0, rather they should be noted as missing data
-the analysis of proteins before and after disease onset in at-risk is noteworthy, however the statistical test used on this appears to be incorrect as these data are paired and the sample size is low. A non-parametric approach is preferred and statistical significance would be unlikely to achieve given the low n.

Reviewer #3

(Remarks to the Author)
I reviewed the machine learning methods of this study with personal interest.

Please find below my points:

Strengths:

- Appropriate use of machine learning techniques: The authors employed LASSO regression for feature selection and logistic regression for building predictive models of treatment response, which are suitable for the study's goals.
- Proper data preprocessing: The authors correctly applied normalization (scale normalization) before using LASSO, which is crucial for this algorithm to perform effectively and without bias towards features with larger scales.
- Robust validation approach: The use of 10-fold cross-validation repeated 100 times provides a stable assessment of model performance and generates reliable ROC curves.
Good predictive performance: The models achieved commendable performance with average ROC AUCs of 0.88-0.95 for training sets and 0.81-0.88 for test sets.
- Independent validation: The authors validated their models on an independent cohort using ELISA measurements of the selected protein biomarkers, which significantly strengthens the reliability of their findings.
- Clear visualization: The presentation of model performance through ROC curves and confusion matrices is clear and informative.

Although I noticed the following limitations:

- Sample size: The relatively small sample sizes, particularly for the MTX+HCQ group (n=59), may limit the robustness of the machine learning models. Larger cohorts would enhance the reliability of the findings.
- Exploration of advanced algorithms: While LASSO is appropriate, the study could benefit from exploring more advanced machine learning techniques like XGBoost or Random Forests. These algorithms can capture non-linear relationships while still handling multicollinearity, potentially improving predictive performance and providing additional insights.
- Hyperparameter tuning: The manuscript lacks details on hyperparameter optimization for the LASSO models. A description of this process would add methodological rigor.
- Class imbalance: More information on how potential class imbalance in response/non-response groups was addressed would be valuable.
- Feature selection robustness: Comparing LASSO with other feature selection methods would help ensure the robustness of the selected biomarkers.
- Model interpretability: Additional details on model interpretability, such as feature importance plots, would provide deeper insights into the most predictive proteins.
- Comparison with clinical predictors (IMPORTANT): A comparison of the machine learning models to prediction rules based on standard clinical variables would contextualize the added value of the proteomic approach.
- Code availability: At the time of this review, the code was not accessible at the provided GitHub repository. Making the code available would enhance reproducibility and allow for a more thorough peer review of the implementation details.

Reviewer #4

(Remarks to the Author)
Overall, the authors have adequately addressed all the points raised by the reviewers. I recommend including a statement in the discussion about the limitations of the relatively small sample sizes in some of the subgroups, including the replication cohort.

Version 2:

Reviewer comments:

Reviewer #5

(Remarks to the Author)
On the comment about limited sample sizes.
- The authors responded satisfactorily. Increasing cohort sizes is clearly out of scope for the current work.

On exploring alternative machine learning techniques.
- The authors responded satisfactorily, including comparisons between LASSO and two well established ensemble methods (RF and XGBoost) and modifying the manuscript accordingly. It is very interesting to see that the simpler approach rendered better results (not unexpected, given data set size limitations).

On hyper parameter tuning.
- The authors responded satisfactorily. The requested information was included and reads clearly.

On class imbalance.
- The authors responded satisfactorily.

On feature selection stability.
- The authors included in this revised version feature selection capabilities the ensemble methods (RF and XGBoost) provide. The response was mostly focused on performance, however the original question was about stability of the selected biomarkers (i.e., do different feature selection approaches lead to the same biomarkers being selected?). I suggest reframing the answer to this remark accordingly.

On model interpretability/explainability
- The authors' response was about hyper parameter optimisation, not model interpretability. I suggest adding feature importance analysis (e.g., SHAP/SHAPley) or other post-hoc explainability tools.

On comparison with clinical predictors
- The authors responded satisfactorily.

On code availability
- The authors responded satisfactorily.

(Remarks on code availability)

Point-by-point responses to reviewer's comments

Reviewers' comments:

Reviewer #1 (Remarks to the Author):

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This is an interesting proteomics study from He et al. which uses a well-described cohort of RA patients, at-risk individuals, and healthy controls, to better understand plasma protein changes associated with the various states. They main results suggests that there are overlapping differences in the plasma proteome between RA and at-risk individuals, proteins that correlate with disease activity, and proteins that predict treatment response with conventional DMARDs, some of which change in individuals who respond to the drug in follow up testing.

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The main drawback of the paper is that no validation methods are deployed to ensure that accessible methodology (ELISA, or other antibody-based methods) is in agreement with these results. There are no mechanistic studies and no external validation cohort, which limits the generalizability of the results.

The methods are well described, the work is original and could be a significant resource for follow up studies.

Response: We greatly appreciate your efforts to evaluate our work. While giving positive feedback, you also provided us with many valuable suggestions. Based on your recommendations, we have thoroughly revised the manuscript. To address the suggestion to validate our findings in an independent cohort, we recruited an additional cohort comprising 46 RA patients receiving MTX+HCQ and 19 RA patients receiving MTX+LEF. We conducted an ELISA analysis to assess the levels of protein biomarkers in this cohort. The results were consistent with our initial findings, confirming the reliability of our predictive models. The validation results from this independent cohort are presented in Fig. 5k and Extended Data Fig. 5e of the revised manuscript, with a detailed description provided in lines 282-291.

Major

-the title speaks to a longitudinal study, with reference to pre-RA. While it's extremely valuable to consider the pre-RA state, these samples are analyzed in cross-section, not longitudinally (ie. the individuals have not progressed into inflammatory arthritis). A change in the title is required.

Response: Thank you for your constructive suggestion. To address this issue, we have now included the missing longitudinal data for individuals at risk for RA. Among the 60 at-risk individuals for RA, 38 were followed up for 5-7 years. The follow-up information is included in revised Supplementary Table 1b. Moreover, we compared the proteomic differences between individuals who progressed to RA and those who did not. The detailed findings are provided in Fig. 2d-f of revised manuscript and described in lines 148-158. Since we have completed the follow-up information for individuals at risk for RA, and our RA cohort also includes follow-up data, we have decided to retain the original title. Thank you for your suggestions, which have greatly improved our manuscript.

-how many of the ACPA+ at-risk have developed RA? An analysis that aims to distinguish those who progress vs those who have not progressed would be valuable, if the numbers were high enough.

Response: We sincerely appreciate your great question. Among the 60 at-risk individuals for RA, 38 were followed up for 5-7 years. The follow-up information is included in revised Supplementary Table 1b. We found that 7 out of 38 followed-up persons have already developed RA, according to 2010 ACR/EULAR criteria of RA. Therefore, following your suggestion, we performed comparative proteomic analysis between at-risk patients who progressed into RA and those who did not. The at-risk patients who progressed into RA showed significantly lower complement components, indicating depletion due to the formation of immune complexes during their transition to RA. Meanwhile, metabolism linked proteins like PSMB7 increased, suggesting immunoproteasome activation (revised Fig. 2d,e and Supplementary Table 2e,f, with a detailed description provided in lines 148-153.).

Additionally, for 3 out of the 7 at-risk patients who progressed to RA, we collected serum samples before and after disease onset and conducted a comparative proteomic analysis. We identified three proteins, APOE, HIST2H3A, and TF, that were consistently detected both between individuals who transitioned to RA and those who did not, as well as before and after RA onset. These findings highlight the critical roles of lipid metabolism dysregulation, neutrophil extracellular trap formation, and iron homeostasis in the development of RA (revised Fig. 2f and Supplementary Table 2g, with a detailed description provided in lines 153-158.).

Thank you for your great suggestions, which have provided us with a deeper understanding of plasma protein changes during RA progression.

-more details pertaining to the ACPA+ at-risk are required. Are there clinical data such as joint pain and stiffness? VAS? What are the ACPA levels of these individuals?

Response: We sincerely appreciate your great questions. Since the at-risk individuals for RA did not show any signs or symptoms of RA, there is no information on joint pain, stiffness, or VAS (Visual Analog Scale) scores. To clarify this, we have added the VAS scores of 0 for joint pain and stiffness to Supplementary Table 1a,b of the revised manuscript. Additionally, we have included the ACPA levels for these ACPA+ at-risk individuals in revised Supplementary Table 1a.

-Figure 1 lays out the experimental design well but all of the quality control figures should be considered for supplemental data

Response: Thanks for your valuable suggestion. We have moved all the quality control figures into revised Extended Data Fig. 1a-c.

-most TLRs are either expressed intracellularly on the plasma membrane, yet these are upregulated in individuals with RA - where are they coming from?

Response: We apologize for the unclear description. To clarify, the TLRs mentioned in the manuscript refer to the enriched pathways rather than the proteins themselves (revised Supplementary Table 2d). The enriched TLR pathways involve S100A12, UBA52, UBE2D2, and UBE2N. According to the Gene Ontology Cellular Component (GO CC) classification, S100A12 and UBA52 are localized in the extracellular region. In contrast, UBE2D2 and UBE2N are associated with the ubiquitin ligase complex and can be secreted into the serum through extracellular exosomes (Supplementary Table 2d).

-the ACPA+ RA patients had higher disease activity than the ACPA- RA patients - is this the primary driver of the inflammatory response signatures or was disease activity controlled for?

Response: We sincerely appreciate your great question. Our findings showed that the ACPA-positive individuals showed lower DAS28-CRP scores and disease activity than the ACPA-negative

RA patients (revised Supplementary Table 1a). Following the reviewer's question, we compared the enrichment scores of inflammatory response pathway between the ACPA-positive and ACPA-negative RA patients, and found that even after adjusting for DAS28-CRP, the scores of the inflammatory response pathway remained significantly higher in ACPA-positive RA patients compared to ACPA-negative RA patients (revised Extended Data Fig. 1e), suggesting that ACPA status may independently contribute to the heightened inflammatory response. However, more evidence is further required for this conclusion.

-the analysis on disease activity trends with age is problematic. 1. CRP tends to increase with age, which is an interesting phenomenon but may increase DAS28-CRP levels for different reasons outside of RA activity. 2. Tender joint counts may be a reflection of joint damage, rather than active joint swelling. I agree however with controlling for age and sex, as was done in the next analysis.

Response: Thank you very much for your insightful questions and valuable suggestions. We assume that the increase in CRP levels with age may contribute to the elevated DAS28-CRP scores independently of RA activity. To address this question, we examined the associations between age and each parameter (CRP, TJC, SJC and VAS) of the DAS28-CRP separately. Interestingly, our findings clearly showed that both CRP and TJC are significantly correlated with age, suggesting a direct link between aging and certain arthritis symptoms (revised Fig. 3f and Extended Data Fig. 2a). The relevant description was shown in lines 172-175. Moreover, we would like to clarify that tender joint counts are clinical indicator of active joint swelling in this work. However, we also agree with you that tender joint counts may also reflect joint damage. Based on the above results, we adjusted for age and sex during all the analyses of disease activity-associated plasma proteins (revised Figs. 3g-j, 4c-f and Extended Data Fig. 2b-f).

-the non-linear analysis is of interest, but the 2 methods employed seem overlapping and somewhat redundant. The agreement between DE-SWAN and the clinically important cut offs for DAS28 are reassuring, however this seems like an overly exhaustive analysis without a specific direction.

Response: Thank you for your interest in our non-linear analysis. The unsupervised clustering and DE-SWAN analyses employed in our study operate on different algorithmic principles and address distinct aspects of the data. Unsupervised clustering analysis focuses on identifying patterns based on similarities in changing trends among groups classified by disease activity. This approach helps to elucidate proteomic mechanisms in relation to inherent disease activity stratification within a clinical context. In contrast, DE-SWAN analysis is specifically designed to detect and account for non-linear associations with high precision. It examines proteomic differences between samples as the DAS28 score shifts in 0.1 increments, potentially revealing important mechanisms that might be missed in broader group comparisons. Although similar results may emerge from both methods, they are complementary. Unsupervised clustering provides clinical subtyping from one perspective, while DE-SWAN offers a detailed examination from another. Together, these methods contribute to building connections between phenotype and plasma protein changes (revised Fig. 4a, 4c-f, and Extended Data Fig. 3). The relevant description has been updated in lines 205-236.

-it's not clear to me why MTX-LEF combo therapy is used in such high rates, is this combination starting together at the first visit, or is LEF being added to MTX if MTX monotherapy is not successful. It is widely known that MTX-LEF is a challenging combination of treatments to use in RA due to their effects on the liver.

Response: Thank you for your critical and insightful question regarding the use of MTX-LEF combination therapy. Due to the historically high cost of biological agents (bDMARDs), it was not uncommon several years ago to initiate treatment with MTX monotherapy or MTX-LEF combination therapy, particularly for patients without full insurance coverage, to achieve earlier low disease activity or remission. This explains why LEF has been widely used in mainland China, as previously reported (Int J Rheum Dis. 2020; 23(12):1719-1727; Clin Rheumatol. 2018; 37(3):597-605). Fortunately, EULAR has recommended this approach as a cost-effective alternative, especially in developing countries where the economic burden of bDMARDs is significant.

We are aware of the potential hepatotoxic effects associated with the MTX-LEF combination. In a previous study, approximately 22% of patients experienced at least one adverse event during LEF treatment. The most frequently reported were gastrointestinal events (41.27%), followed by elevated liver enzyme levels (19.74%) (Clin Rheumatol. 2018; 37(3):597-605). To reduce the risk of hepatic injury, we employ monthly blood tests to monitor liver function during treatment, both in daily practice and within our cohort. If any signs of liver damage are detected, we promptly adjust the csDMARDs regimen or advise the patient to consider biological agents.

-The results on predicting treatment responses are appealing, due to the high ROC and low # of inputs required. Can these proteins be measured using multiple reaction monitoring or ELISA as a follow up?

Response: Thank you for your valuable suggestion. Based on your recommendation, we validated our findings in an independent cohort consisting of 19 RA patients receiving MTX+LEF treatment and 36 RA patients receiving MTX+HCQ treatment. We conducted a three-month follow-up to monitor their treatment responses. Within this cohort, we measured the levels of the selected biomarkers using ELISA. The results confirmed that these biomarkers effectively predicted treatment responses, consistent with our initial findings (revised Fig. 5k and Extended Data Fig. 5e).

The relevant description has been updated in lines 283-291.

Minor

-line 52: “at-risk” includes other individuals, namely first-degree relatives, individuals carrying shared epitope, etc.

Response: Thank you for your careful reminder regarding the definition of "at-risk" individuals. We have enhanced the text by providing a more detailed definition of "at-risk" throughout all phases of at-risk RA, following the criteria set by the ACR/EULAR (Ann Rheum Dis. 2012; 71(5):638-641). Additionally, we clarified that in our study, the term "at-risk" specifically refers to ACPA-positive individuals who do not exhibit any symptoms or signs of RA. This aligns with the EULAR category of systemic autoimmunity associated with RA. The revised textual description is provided in the Supplementary Information (lines 34-39).

-line 116: does this reference show that neuronal expansion pathways are upregulated in at-risk individuals?

Response: Thank you for bringing this to our attention. We apologize for the ambiguity caused by the reference. The cited reference indeed indicates the upregulation of axon guidance-related genes in individuals with RA compared to healthy controls, but it does not specifically address the upregulation of neuronal expansion pathways in at-risk individuals. Following the suggestion of Reviewer#2, we reanalyzed the proteome data across clinical subtypes and different sex groups. The neuronal expansion pathways were not among the top-ranked in these analyses. As a result, we have revised the text description and removed this reference.

-line 131: gender or sex?

Response: Thank you for your kind reminder. We used "sex" instead of "gender in the revised manuscript.

Reviewer #2 (Remarks to the Author):

• The manuscript by He et al, investigates the potential of identifying plasma biomarkers of Disease response and possibly pre-disease onset. Predicting disease onset and response is one of the key unmet needs in the field of Rheumatology.

• However, I do have a few queries.

• There is not enough information on how the authors defined individuals at risk. They state ACPA+ and aches/pains with no clinical sign of disease activity. However, they need to be more precise in their exact description, IAR should be categorised according to EULAR criteria where clinical assessment of these individuals gives a score <6, thus not reaching criteria for definitive RA. They should include any clinical details that were assessed, is CRP, SJC, TJC, VAS. Also did they use ultrasound guided analysis to define definitively that there was no joint inflammation. One or both of these approaches are the goal standard for defining IAR.

Response: Thank you for your insightful questions and valuable suggestions. We apologize for not providing complete basic clinical information for the individuals at-risk (IAR) in our initial submission. We have now included detailed information about the ACPA levels for each IAR (Supplementary Table S1a). In alignment with the EULAR criteria, we categorized at risk individuals as those in the "systemic autoimmunity associated with RA" phase. Specifically, we selected ACPA-positive individuals who do not exhibit any symptoms or clinical signs of RA. Consequently, all Tender Joint Count (TJC), Swollen Joint Count (SJC), and Visual Analog Scale (VAS) scores for these individuals were zero, indicating no clinical sign of arthritis. We did not use ultrasound-guided analysis in this study to definitively rule out joint inflammation, but we acknowledge that this is a gold standard approach for defining IAR. The definition of IAR was supplemented in the "Study design and ethics approval" part of revised Supplementary information (lines 34-39).

• The authors do discuss effect of age and gender. The IAR group is much younger than the RA and HC groups, did they take this into account. They should do a sub-analysis comparing aged-matched HC, RA and IAR groups.

Response: Thank you for your constructive suggestion regarding the potential effects of age and gender on our analysis. Our findings revealed that the median age of the at-risk (IAR) group is slightly lower than that of the RA and healthy control (HC) groups, with an age difference of only about 1 year. Additionally, this difference was not statistically significant (revised Extended Data Fig. 1g). Given the lack of significant age differences between the HC, RA, and IAR groups, we decided not to conduct a sub-analysis based on age.

• The authors also discuss differences in gender in the RA group. The % females in the IAR group are different from RA group (60% vs 85%) therefore they should perform a sub-analysis.

Response: We sincerely appreciate your valuable suggestion. Following your recommendation, we conducted a gender-specific analysis, and the results have been incorporated into the revised manuscript (revised Extended Data Fig. 1d, 1h). The relevant description has been updated in lines 137-147.

• For Figure 2, they show volcano blots comparing two groups at a time, it would be more useful to perform unsupervised hierarchical clustering for all 4 groups to see if they separate out into distinct phenotype groups based on plasma proteins or is there overlap.

Response: Thank you for your constructive suggestions. Based on your recommendation, we replaced the volcano plot with an unsupervised clustering analysis (revised Fig. 2b). Interestingly, the hierarchical clustering revealed clear distinctions between the clinical groups, with at-risk RA and ACPA+ RA clustering more closely together. These findings have been incorporated into the revised manuscript in lines 104-105.

• Figure 2d, are the authors surprised there is a divergence in the level of some proteins between IAR and RA, would they not expect that as IAR is pre-disease onset they might just all follow stepwise progression from HC.

Response: We appreciate your insightful question regarding the divergence in protein levels between the IAR and RA groups. Our clustering results indicate that the at-risk RA is actually closer to ACPA-positive RA in disease progression than to healthy individuals. This suggests that, despite the absence of symptoms, increased blood inflammation in the IAR group indicates that internal changes have already begun to trend toward RA (Fig. 2b in the revised manuscript).

Indeed, we also observed that some proteins exhibit the highest or lowest levels in the at-risk RA group compared to both the RA and HC groups. While at-risk RA is clinically considered an intermediate stage between HC and RA, we believe that IAR represents a distinct biological stage with its own unique protein expression profile, rather than just an intermediate step. These divergent proteins may serve as biomarkers for early detection or as targets for therapeutic intervention, offering opportunities to alter the disease course before the onset of clinical RA (Fig. 2c in the revised manuscript).

• In Figure 2d, they show HC vs IAR vs all RA, then in Figure 2f and g they compare HC vs RA ACPA-neg vs RA ACPA-pos. Considering IAR are also positive, inclusion of this cohort is important for this figure to investigate if IAR differ against ACPA-neg and ACPA-pos.

Response: Thank you for your constructive suggestion. Following your recommendation, we explored whether IAR differs from ACPA-negative and ACPA-positive RA. Our findings suggest that at-risk RA is more similar to ACPA-positive RA and shows greater differences from ACPA-negative RA (revised Fig. 2b). Specifically, we found that in ACPA-positive RA, signaling pathways such as neutrophil degranulation, biological oxidations, innate immune system, immune system, and acute-phase response were more active compared to at-risk RA. In contrast, pathways related to lipid metabolism, fatty acid degradation, and XBP1(S) activation of chaperone genes were more active in ACPA-negative RA (revised Fig. 2c). Our findings clearly demonstrate the plasma proteome disparities between at-risk RA, ACPA-positive RA, and ACPA-negative RA.

• Figure 3- focuses on gender, and observe differences, which is expected but nice to see. Again, here based on the differences in % gender in IAR group, analysis of this group would be important to examine in this context.

Response: Thank you for your constructive suggestions. Given the varying gender distribution in the IAR group, analyzing the sex effect within this context is indeed important. Following your recommendation, we performed a sex-specific analysis and found that most differentially expressed proteins (DEPs) remained consistent both before and after sex stratification (revised Extended Data Fig. 1d,h), indicating that sex differences have minimal effects on plasma protein changes, despite the differences in disease activity between females and males. Interestingly, we observed that the enriched pathways using DEPs between responders and non-responders following MTX+LEF treatment differed between females and males (revised Extended Data Fig. 5a-d), though the mechanism behind this phenomenon is unclear. Additionally, we compared the proteome differences between samples before and after treatment in male and female responders and non-responders (revised Extended Data Fig. 6c-g). The relevant descriptions have been updated in lines 137-147, lines 257-268 and lines 310-319.

• Did they expect the DAs28 to be higher in males, it tends to be more associated with females? Were correlations maintained if they split the group into ACPA Pos and ACPA-neg and separately analysed.

Response: Thank you for your insightful question. A previous study elegantly demonstrated that DAS28 is slightly higher in females than in males in early RA (Ann Rheum Dis. 2010;69(1):230-3), and we have cited this reference accordingly. However, our cohort showed a different result, with higher DAS28 scores in males (Fig. 3a of the initial submission). Following your recommendation, we further analyzed the differences by splitting the group into ACPA-positive and ACPA-negative subgroups to determine if these correlations are maintained. We found that the higher DAS28 in males was observed only in ACPA-positive RA, not in ACPA-negative RA (Fig. 3a,b of the revised manuscript). Additionally, we observed that DAS28 increased with age in both male and female ACPA-positive RA patients (Fig. 3c of the revised manuscript). In contrast, the changes in DAS28 with age differed between males and females in ACPA-negative RA (Fig. 3d of the revised manuscript). The relevant descriptions have been updated in lines 165-168.

• In figure 4 they show 6 specific gene clusters. Then perform unsupervised hierarchical clustering which didn't actually show specific separation. When they perform a wave DAS 28 clustering the authors suggest there are three peaks 2.9, 4.0 and 5.1. However, the clustering in the heatmap does not seem that definitive. What do they think the significance of these three peaks are regarding clinical practice and assessment.

Response: Thank you for your insightful question. The unsupervised hierarchical clustering revealed distinct trends in protein changes across DAS28 clusters. Specifically, clusters 5 and 6 show a general decrease in protein levels as disease activity increases, with cluster 5 declining after low disease activity and cluster 6 decreasing from the beginning. Cluster 3 gradually increases, cluster 4 decreases initially before sharply rising at moderate activity levels, cluster 2 rises early then gradually decreases after low disease activity, and cluster 1 exhibits a fluctuating pattern. Notably, our analysis highlighted differences in DAS28-related proteomic variations between ACPA-positive and ACPA-negative groups, prompting separate clustering analyses for each (revised Fig. 4a and Extended Data Fig.3). The revised results have been updated in lines 209-222.

To address the concern regarding the less distinct appearance of DE-SWAN peaks in the LOESS heatmap, we clarify that the LOESS plot reflects protein levels after applying a smoothing algorithm, resulting in a more generalized trend when plotted against the DAS28 axis. In contrast, DE-SWAN analysis focuses on identifying significant proteins at each DAS28 point (0.1 intervals in our analysis), with the three peaks selected based on the highest number of significantly altered proteins. Consequently, the differences highlighted by DE-SWAN may not be as visually prominent in the LOESS plot. Both approaches have been used similarly in previous proteomics studies (Nat Med. 2019;25(12):1843-1850).

Due to the smaller number of ACPA-negative RA patients, the revised manuscript includes DE-SWAN analysis only for ACPA-positive patients, leading to slight shifts in the identified peaks (from 2.9 to 3.1, 4.0 to 3.8, and 5.1 to 5.0) (revised Fig. 4c). These peaks, close to DAS28-CRP thresholds for moderate disease activity (3.2-5.1), provide valuable treatment insights. Proteins altered at 3.1 and 3.8 suggest prioritizing anti-inflammatory treatments in low to moderate activity, while those at 5.0 emphasize the need to address immune-metabolic dysregulation in high disease activity (revised Fig. 4e). The relevant descriptions have been updated in lines 225-236.

• Figure 5-6, identify proteins associated with response. Firstly, it would be interesting to analyse ACPA pos and ACPA neg separately, as it is thought they respond differently. Secondly based on their gender data, analysis of female vs males' responses would also be important, as it is known males respond better.

Response: Thank you for your thoughtful and great recommendations. Following your suggestions, we carried out the difference analysis based on ACPA status and sex.

We first analyzed the differences between responders and non-responders in ACPA-positive RA. In the MTX+LEF group, responders exhibited elevated complement activation, fibrinolysis, and autophagy, alongside downregulated metabolic and glycolytic pathways. In the MTX+HCQ group, immune activation was upregulated, while mitochondrial transport pathways were downregulated in ACPA-positive responders (revised Extended Data Fig. 5a-d). Due to the small sample size of responders and non-responders in the ACPA-negative RA group, further analysis of this subgroup was not conducted. Additionally, we analyzed the differences between responders and non-responders separately in females and males. In female responders to MTX+LEF treatment, protein transport and inflammatory pathways were elevated, whereas male responders showed increased endocytosis (revised Extended Data Fig. 5a,b and Supplementary Table 5e,f). For the MTX+HCQ group, the male sample size was too small for meaningful analysis, so the analysis was conducted exclusively in females, revealing increased nonsense-mediated decay and decreased amino acid metabolism in responders (revised Extended Data Fig. 5c,d and Supplementary Table 5g,h). The relevant descriptions have been updated in the manuscript in lines 257-268.

Due to the small sample size of ACPA-negative RA patients, we analyzed protein differences before and after treatment only in ACPA-positive RA patients. After MTX+LEF treatment, ACPA-positive RA patients exhibited upregulation in translation, amino acid metabolism, and axon guidance pathways. In the MTX+HCQ-treated group, RNA metabolism, neutrophil degranulation, and axon guidance were elevated, while complement activation was reduced (revised Extended Data Fig. 6c-g and Supplementary Table 6d,f). Additionally, we considered the effects of sex on these proteomic differences (revised Extended Data Fig. 6c-g and Supplementary Table 6d,f). The relevant descriptions have been updated in the manuscript in lines 310-319.

Finally, where the paper is lacking is the further analysis of the IAR group throughout. In particular, how many of the IAR patients converted to RA vs non-convertors. This really is the key question. Had convertors higher levels of protein, did those that converted earlier have higher levels. Do the authors have plasma from those when they converted vs those that did not.

Response: We sincerely appreciate your great question. Among the 60 at-risk individuals for RA, 38 were followed up for 5-7 years. The follow-up information is included in revised Supplementary Table 1b. We found that 7 out of 38 followed-up persons have already developed RA, according to 2010 ACR/EULAR criteria of RA. Therefore, following your suggestion, we performed comparative proteomic analysis between at-risk patients who progressed into RA and those who did not. The at-risk patients who progressed into RA showed significantly lower complement components, indicating depletion due to the formation of immune complexes during their transition to RA. Meanwhile, metabolism linked proteins like PSMB7 increased, suggesting immunoproteasome activation (revised Fig. 2d,e and Supplementary Table 2e,f, with a detailed description provided in lines 148-153.).

Additionally, for 3 out of the 7 at-risk patients who progressed to RA, we collected serum samples before and after disease onset and conducted a comparative proteomic analysis. We identified three proteins, APOE, HIST2H3A, and TF, that were consistently detected both between individuals who transitioned to RA and those who did not, as well as before and after RA onset. These findings highlight the critical roles of lipid metabolism dysregulation, neutrophil extracellular trap formation, and iron homeostasis in the development of RA (revised Fig. 2f and Supplementary Table 2g, with a detailed description provided in lines 153-158.).

Thank you for your great suggestions, which have provided us with a deeper understanding of plasma protein changes during RA progression.

• The authors are suggesting the markers identified are potential protein biomarkers for RA. Do they think they are specific to RA, i.e., are these markers generally up in inflammatory diseases, i.e., other rheumatic diseases like PsA, Ank Spon, outside the field like inflammatory bowel disease, psoriasis. Think this would be useful.

Response: Thank you for your excellent question. In response to the request for validation that our findings are specific to RA and not to other rheumatic or inflammatory diseases, we conducted a comparative analysis of high-abundance protein biomarkers in RA versus other inflammatory diseases, including primary Sjögren's syndrome, systemic sclerosis, idiopathic inflammatory myopathy, and systemic lupus erythematosus. Our results demonstrate that most of these proteins are specifically elevated in RA, rather than being associated with other inflammatory conditions. These findings are presented in Extended Data Fig. 1f of the revised manuscript, with a detailed description provided in lines 131-136.

• Would they use one protein or a panel of proteins?

Response: We apologize for the unclear description. In our models for predicting treatment response, a panel of proteins was used to construct models for each therapeutic strategy (Fig. 5i-j). This same set of proteins was also applied for independent validation using another cohort (revised Fig.5k and Extended Data Fig. 5e). The relevant description has been updated in lines 282-291 of the revised manuscript and in the Methods section of the Supplementary Information (lines 166-191).

• They would need to validate the panel in an additional cohort.

Response: Thank you for your valuable suggestion. Based on your recommendation, we validated our findings in an independent cohort consisting of 19 RA patients receiving MTX+LEF treatment and 36 RA patients receiving MTX+HCQ treatment. We conducted a three-month follow-up to monitor their treatment responses. Within this cohort, we measured the levels of the selected biomarkers using ELISA. The results confirmed that these biomarkers effectively predicted treatment responses, consistent with our initial findings (revised Fig. 5k and Extended Data Fig. 5e). The relevant description has been updated in lines 282-291.

NCOMMS-24-17994A

Point-by-point response to reviewers' comments.

Reviewers' comments:

Reviewer #1 (Remarks to the Author):

The authors have undertaken a significant revision of their manuscript and most of my critiques have been addressed.

I only have one minor comment:

-if the ACPA+ at-risk individuals were not assessed for tender joints or pain VAS, it is not appropriate to score these as 0, rather they should be noted as missing data

Response: Thank you for your professional suggestion. We have revised the clinical data for SJC, TJC, VAS and CRP following your suggestion (please see revised Supplementary Table 1a).

-the analysis of proteins before and after disease onset in at-risk is noteworthy, however the statistical test used on this appears to be incorrect as these data are paired and the sample size is low. A non-parametric approach is preferred and statistical significance would be unlikely to achieve given the low n.

Response: Thank you for your constructive suggestion. Regarding the statistical approach used, we would like to clarify the following:

Firstly, we performed normality tests, and the majority of the proteins demonstrated a normal distribution before and after disease onset (Shapiro test, p > 0.05), supporting the use of parametric methods (**FIGURE 1A**). As a result, the 10 proteins significantly different between before and after disease onset groups exhibited consistent trends in all patients, further validating the reliability of our findings (**FIGURE 1B**).

We agree with that non-parametric methods are preferred in cases of small sample sizes. Following your suggestion, we also performed non-parametric test to screen the differential proteins. However, no differential proteins were obtained. We assume that when the sample size is below five, calculating differences using the U-test can be difficult. Notably, several previous studies have also used the parametric tests with sample sizes as small as three (***Nat Commun.*** 2025; 16(1):524; ***Nat Commun.*** 2025;16(1):598). Given the normal distribution of the proteomic data before and after disease onset, and the results from parametric and non-parametric tests, we preferred to use the parametric tests in this data analysis.

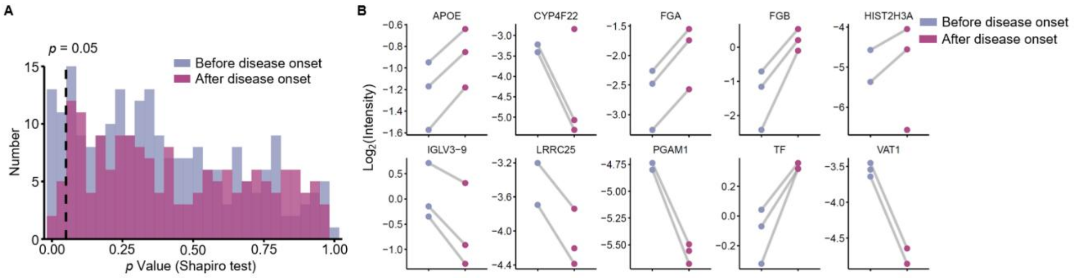

FIGURE 1. Screening of differential proteins before and after disease onset.

A. Bar chart showing the normality test results for plasma proteins before and after disease onset (Shapiro test).

B. Scatter plot with a line graph illustrating the changes in 10 differential proteins before and after disease onset in three patients.

Reviewer #3 (Remarks to the Author):

I reviewed the machine learning methods of this study with personal interest. Please find below my points:

Strengths:

- Appropriate use of machine learning techniques: The authors employed LASSO regression for feature selection and logistic regression for building predictive models of treatment response, which are suitable for the study's goals.

- Proper data preprocessing: The authors correctly applied normalization (scale normalization) before using LASSO, which is crucial for this algorithm to perform effectively and without bias towards features with larger scales.

- Robust validation approach: The use of 10-fold cross-validation repeated 100 times provides a stable assessment of model performance and generates reliable ROC curves.

Good predictive performance: The models achieved commendable performance with average ROC AUCs of 0.88-0.95 for training sets and 0.81-0.88 for test sets.

- Independent validation: The authors validated their models on an independent cohort using ELISA

measurements of the selected protein biomarkers, which significantly strengthens the reliability of their findings.

- Clear visualization: The presentation of model performance through ROC curves and confusion matrices is clear and informative.

Response: We greatly appreciate the reviewer's positive comments on our work.

Although I noticed the following limitations:

- Sample size: The relatively small sample sizes, particularly for the MTX+HCQ group (n=59), may limit the robustness of the machine learning models. Larger cohorts would enhance the reliability of the findings.

Response: We appreciate the reviewer's insightful comments. We acknowledge that the sample sizes, particularly for the MTX+HCQ group, are relatively small. The challenges in recruiting a sufficient cohort for the drug response prediction model stem from the stringent requirements for enrolling drug-naïve subjects or those who have not received csDMARDs treatment for at least 6 months prior to the collection of plasma samples.

To mitigate the limitations posed by small sample sizes, we conducted 100 rounds of resampling to enhance model stability (lines 282-284 in the revised manuscript). Furthermore, we validated our model using an independent cohort of 46 patients treated with MTX+HCQ and 19 patients treated with MTX+LEF. ELISA results from this validation cohort showed consistent biomarker changes with the proteomic data between responders and non-responders. Incorporating these protein levels into our model maintained strong classification performance, with ROC values of 0.90 for MTX+LEF treatment and 0.86 for MTX+HCQ treatment. These validation results underscore the reliability and robustness of our models. The relevant descriptions were in lines 302-313 in the revised manuscript.

We agree with the reviewer that larger cohorts would strengthen the reliability of our findings, however, recruiting such cohorts in a short period remains challenging. We hope to expand the sample sizes in future studies to further improve the model's robustness.

- Exploration of advanced algorithms: While LASSO is appropriate, the study could benefit from exploring more advanced machine learning techniques like XGBoost or Random Forests. These algorithms can capture non-linear relationships while still handling multicollinearity, potentially improving predictive performance and providing additional insights.

Response: We sincerely appreciate your great suggestion. Following your recommendation, we applied both XGBoost and Random Forests to generate prediction models using the feature proteins obtained from LASSO regression. For Random Forest, we used the following settings, including 500 trees (ntree = 500), the square root of the number of features for splits (mtry), and a minimum node size of 1. For XGBoost, we set the maximum tree depth to 4, learning rate to 1, 10 boosting rounds, and 2 threads. For the MTX+LEF treatment group, the ROC values obtained using Random Forests ranged from 0.55 to 0.96, with a median ROC of 0.79, while XGBoost yielded ROC values ranging from 0.50 to 0.95, with a median of 0.73. These results were slightly lower than those achieved with LASSO regression, which ranged from 0.63 to 1.00, with a median ROC of 0.88 (**FIGURE 2A**). Similarly, for the MTX+HCQ treatment group, Random Forests achieved ROC values ranging from 0.41 to 0.98, with a median of 0.79, while XGBoost produced values ranging from 0.45 to 0.98, with a median of 0.76. Again, these were slightly lower than LASSO regression, which ranged from 0.53 to 0.95, with a median of 0.82 (**FIGURE 2B**).

Compared with XGBoost and Random Forests, LASSO regression provided more robust prediction models in this study (**FIGURE 2C**). We assume that the relatively small sample size likely constrains the stability and effectiveness of XGBoost and Random Forests, as these methods are more prone to overfitting in small cohorts (***J Clin Med***. 2021;10(19):4393.). The relevant descriptions were in lines 290-298 in the revised manuscript.

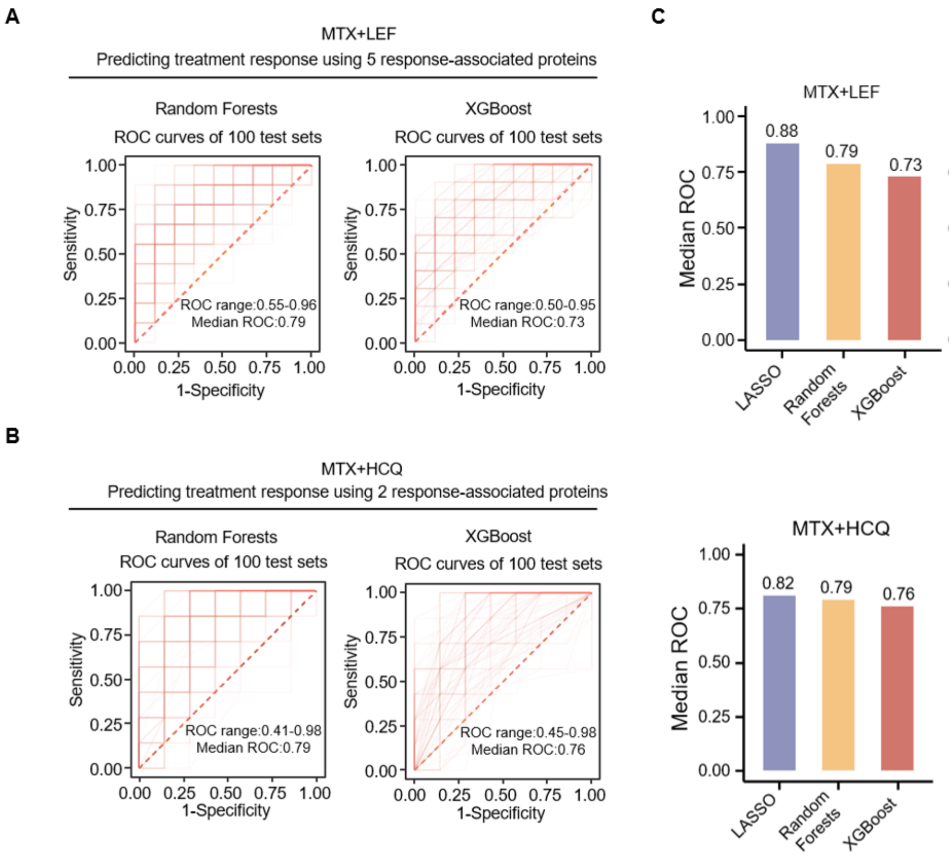

FIGURE 2. Machine learning for predicting treatment response to csDMARDs using XGBoost and Random Forests.

A. ROC curves illustrating the performance of the Random Forest (left) and XGBoost (right) models for predicting the treatment response to MTX+LEF, based on 5 proteins features selected by LASSO. This process was repeated 100 times (refer to Extended Data Fig. 6c in the revised manuscript).

B. ROC curves illustrating the performance of the Random Forest (left) and XGBoost (right) models for predicting the treatment response to MTX+HCQ, based on 2 proteins features selected by LASSO. This process was repeated 100 times (refer to Extended Data Fig. 6e in the revised manuscript).

C. Bar chart showing the median ROC values of the three models (Random Forest, XGBoost, and LASSO), using 5 proteins features selected by LASSO for the MTX+LEF (left) and 2 proteins features selected by LASSO for the MTX+HCQ (right) groups (refer to Extended Data Fig. 6f in the revised manuscript).

- Hyperparameter tuning: The manuscript lacks details on hyperparameter optimization for the LASSO models. A description of this process would add methodological rigor.

Response: Thank you for your valuable suggestion. We apologize for not including enough details on hyperparameter optimization for the LASSO models. We have now added "Machine learning for treatment response" in Methods section of the revised manuscript (lines 540-560).

- Class imbalance: More information on how potential class imbalance in response/non-response groups was addressed would be valuable.

Response: Thank you for your valuable suggestion. To address class imbalance, we ensured an equal number of responders and non-responders in each random sampling (refer to Methods section, lines 546-548). After balancing the samples, the median ROC for the MTX+LEF treatment group in the training set increased from 0.95 to 0.96, and the median ROC didn't change in the test set. For the MTX+HCQ treatment group, the median ROC in the training set increased from 0.87 to 0.92, and the median ROC in the test set increased from 0.81 to 0.82 (**FIGURE 3A**). The relevant descriptions were in lines 280-290 in the revised manuscript.

In the validation cohort, after balancing the samples, we used the new models to assess the ELISA results. The results from MTX+LEF treatment group remained unchanged, while the false positive rate in the MTX+HCQ treatment response group decreased from 10.5% to 0% (**FIGURE 3B**). The relevant descriptions were in lines 306-310 in the revised manuscript.

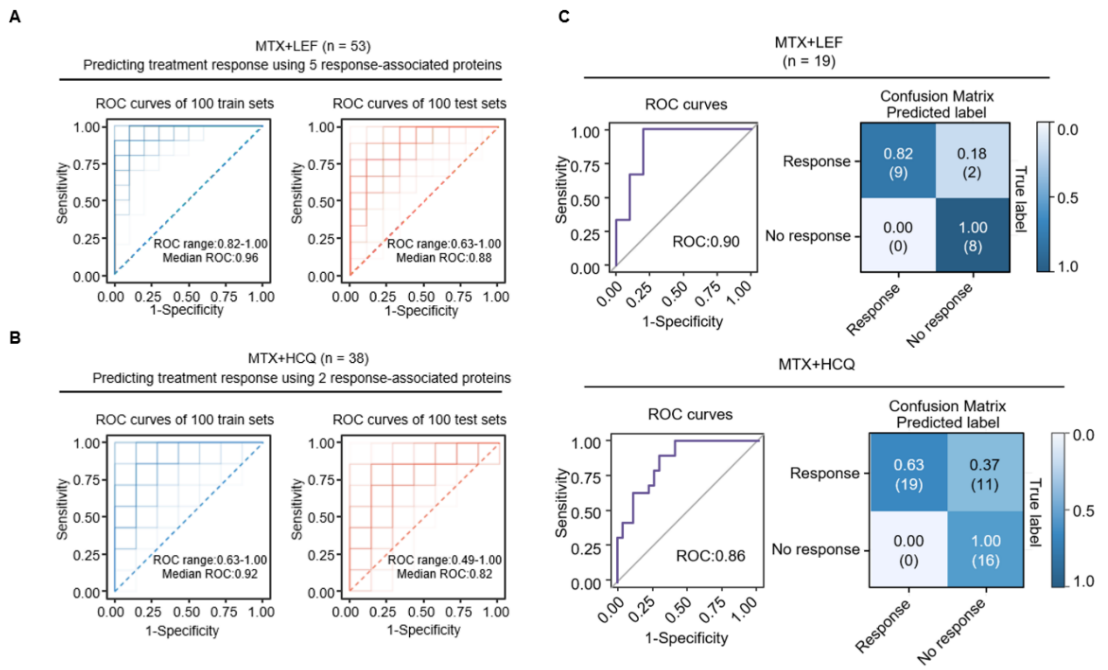

FIGURE 3. Machine learning for predicting treatment response to csDMARDs using LASSO.
A,B. ROC curve illustrating the predictive performance of the LASSO regression model for treatment response to MTX+LEF (**A**) and MTX+HCQ (**B**), utilizing the top 5 or 2 proteins separately. It depicts the effectiveness of the models on the training set (left) and the testing set (right), with a 10-fold cross-validation repeated 100 times (refer to Fig. 5i,j in the revised manuscript).
C. ROC curve indicating the model's performance after applying protein levels measured by ELISA to the response prediction model, with the confusing matrix showing the sensitivity and specificity at the best cutoff given by the ROC curve. The top panel shows the MTX+LEF group, and the bottom panel shows the MTX+HCQ group (refer to Fig. 5k in the revised manuscript).

- Feature selection robustness: Comparing LASSO with other feature selection methods would help ensure the robustness of the selected biomarkers.

Response: Thank you for your valuable suggestion. To address this issue, we applied Random Forest and XGBoost for feature selection, and the prediction models were generated separately based on the protein biomarkers selected by each method. As a result, in the MTX+LEF group, the median ROC values were 0.71 for Random Forests and 0.70 for XGBoost, both of which were lower than the median ROC value of 0.88 for LASSO (**FIGURE 4A**). Similarly, in the MTX+HCQ group, the median ROC values were 0.70 for Random Forests and 0.75 for XGBoost, both of which were lower than the median ROC value of 0.82 for LASSO (**FIGURE 4B**). Collectively, these results highlight the robustness of the LASSO algorithm in selecting features and generating prediction models with strong performance. The relevant descriptions were in lines 290-293 and lines 295-297 in the revised manuscript.

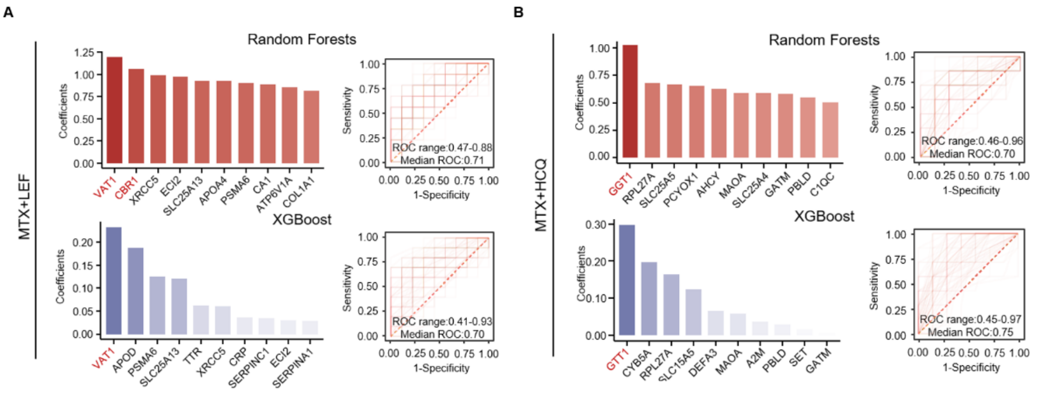

FIGURE 4. Feature selection for predicting treatment response to csDMARDs using XGBoost and Random Forests.

A. Bar plots (left) showing the feature selection results for the Random Forest (top) and XGBoost (bottom) models in the MTX+LEF treatment group. Features selected are marked in red, with thresholds of > 1 for Random Forest and > 0.2 for XGBoost. The ROC curves displaying the prediction performance of both models (right) (refer to Extended Data Fig. 6b in the revised manuscript).

B. Bar plots (left) showing the feature selection results for the Random Forest (top) and XGBoost (bottom) models in the MTX+HCQ treatment group. Features selected are marked in red, with thresholds of > 1 for Random Forest and > 0.2 for XGBoost. The ROC curves displaying the prediction performance of both models (right) (refer to Extended Data Fig. 6d in the revised manuscript).

- Model interpretability: Additional details on model interpretability, such as feature importance plots, would provide deeper insights into the most predictive proteins.

Response: Thank you for your great suggestion. We have added supplementary figures (**FIGURE 5**) detailing the hyperparameter optimization process for the LASSO models. This provides greater methodological rigor and transparency to our approach. We appreciate your valuable suggestion in helping us improve the clarity and robustness of our study. The relevant descriptions were in lines 282-284 in the revised manuscript.

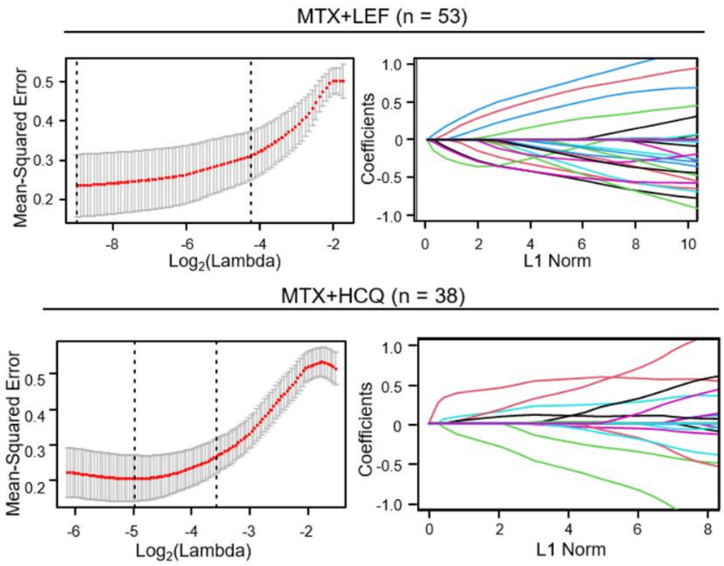

FIGURE 5. The hyperparameter optimization process for the LASSO models in MTX+LEF and MTX+HCQ. For each drug combination, the LASSO path plot (left) shows how the mean squared error changes with the log-transformed penalty coefficient (Log_2(Lambda)), with the red dashed line indicating the optimal lambda value selected based on 10-fold cross-validation. The penalty graph (right) displaying how the coefficients of the selected features change with the L1 norm (x-axis), with different colored lines representing different features. Features with non-zero coefficients at the optimal lambda are considered selected for prediction (refer to Extended Data Fig. 6a in the revised manuscript).

- Comparison with clinical predictors (IMPORTANT): A comparison of the machine learning models to prediction rules based on standard clinical variables would contextualize the added value of the proteomic approach.

Response: Thank you for your insightful suggestion. Based on your suggestion, we incorporated DAS28 parameters (VAS, SJC, TJC and CRP) into the feature set and observed that the predictive performance was slightly improved, with a median ROC of 0.90 compared to 0.88 without using clinical parameters in the MTX+LEF treatment group, and a median ROC of 0.84 compared to 0.82 without using clinical parameters in the MTX+HCQ treatment group (**FIGURE 6**). The relevant descriptions were in lines 298-301 in the revised manuscript.

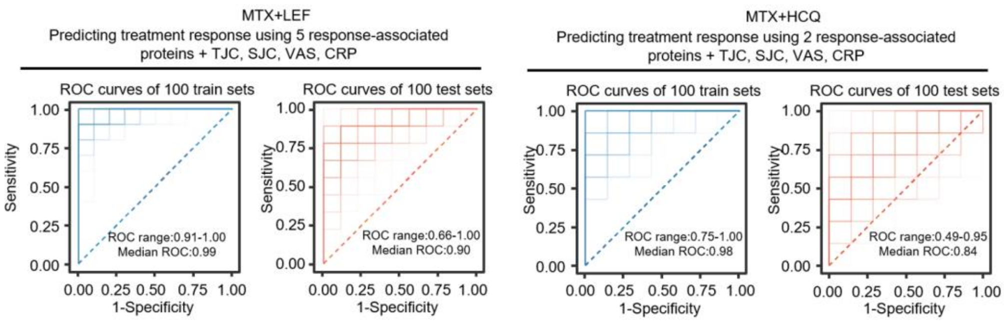

FIGURE 6. ROC curves illustrating the performance of the LASSO model in predicting treatment response to MTX+LEF (left) and MTX+HCQ (right), using the top 5 or 2 proteins selected by LASSO along with four clinical features (SJC, TJC, VAS, and CRP) (refer to Extended Data Fig. 6g in the revised manuscript).

- Code availability: At the time of this review, the code was not accessible at the provided GitHub repository. Making the code available would enhance reproducibility and allow for a more thorough peer review of the implementation details.

Response: We sincerely appreciate the reviewer's valuable suggestion. We have now uploaded the relevant code to GitHub, and it is accessible for review. You can find the code at the following link: https://github.com/hesy1191569605/rheumatoid-arthritis/. We have now added link in Code availability section of the revised manuscript (lines 620-621). Open the link to access individual folders for each figure (Figure 1–6). Each folder contains a "csv" folder for input files, an "outfile" folder for output files, and an R script file titled "Code related to Figure X.R" which includes all relevant code. Additionally, "README.txt" file provides a comprehensive list of the required R packages and environment configurations.

Reviewer #4 (Remarks to the Author):

Overall, the authors have adequately addressed all the points raised by the reviewers. I recommend including a statement in the discussion about the limitations of the relatively small sample sizes in some of the subgroups, including the replication cohort.

Response: Thank you for your positive feedback. Based on your suggestion, we have added a statement in the discussion section to address this limitation, as shown below.

*"While this study provides valuable insights into both the pathogenic mechanisms and pharmacological strategies in RA, it is important to acknowledge several limitations, particularly the relatively small sample sizes in certain subgroups, including those at risk before and after disease onset, as well as in the cohort used to validate the drug response prediction model. The limited sample sizes may be partially attributable to the small number of at-risk individuals who progress to clinical disease. Previous studies have shown that ACPA-positive individuals with arthralgia have approximately a 28% risk of developing RA (**Ann Rheum Dis**. 2011; 70(1):128-33). Although our at-risk subjects were asymptomatic, our follow-up data revealed that 8 out of 38 individuals (21.1%) progressed to RA, reflecting a consistent progression rate. Long-term follow-up (5–7 years) resulted in a limited number of samples available for comparison between converters and non-converters."* The relevant descriptions were in lines 398-407 in the revised manuscript.

NCOMMS-24-17994B

Point-by-point response to the remaining concerns from Reviewer #5

Reviewer #5 (Remarks to the Author):

1. On feature selection stability.

The authors included in this revised version feature selection capabilities the ensemble methods (RF and XGBoost) provide. The response was mostly focused on performance, however the original question was about stability of the selected biomarkers (i.e., do different feature selection approaches lead to the same biomarkers being selected?). I suggest reframing the answer to this remark accordingly.

Response: Thank you for this insightful suggestion. Following your guidance, to directly evaluate the robustness of biomarkers selected by LASSO, we evaluated their rankings across five feature selection methods: Elastic Net, XGBoost, Random Forest, recursive feature elimination combined with support vector machine (REF+SVM), and stability selection. As shown in **FIGURE 1**, for the MTX+LEF group, ECI2 consistently ranked within the top 10 across all methods, while the other four biomarkers (LGALS3BP, CBR1, MYH9, and COL1A1) were ranked within the top 10 in at least three methods. For the MTX+HCQ group, two biomarkers consistently ranked within the top 10 across all methods. These findings support the robustness of the LASSO-selected biomarkers for treatment response. The corresponding depiction has been added to the Results and Methods sections, please refer to lines 268-271and 537-553.

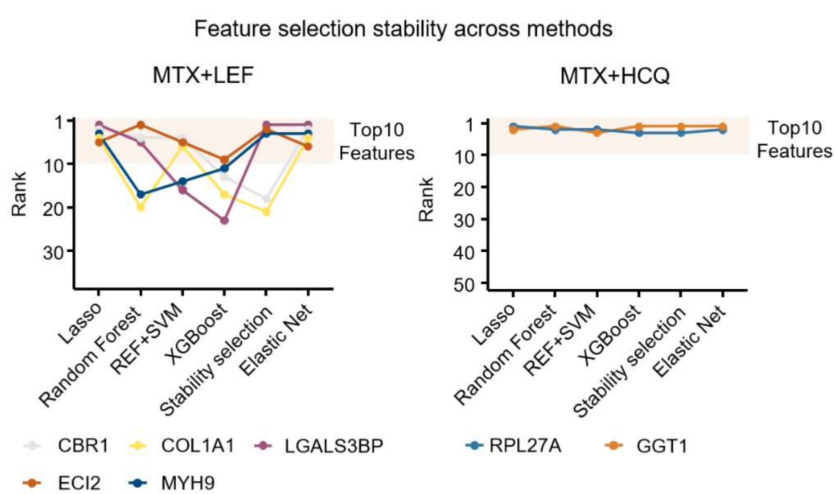

FIGURE 1. Line plots showing the ranking of protein biomarkers for MTX+LEF (left) and MTX+HCQ (right) across LASSO, Random Forest, REF+SVM, XGBoost, stability selection, and Elastic Net. Each colored line represents a single biomarker (refer to Extended Data Figure 6b in the revised manuscript).

2. On model interpretability/explainability

The author's response was about hyper parameter optimisation, not model interpretability. I suggest adding feature-important analysis (e.g., SHAP/SHAPley) or other post-hoc explainability tools.

Response: We appreciate your helpful suggestion regarding model interpretability. In response, we have conducted a SHAP analysis to provide post-hoc interpretability of the models based on the selected biomarkers. As shown in **FIGURE 2**, we visualized the SHAP values for the protein features. For the MTX+LEF group, high expressions of MYH9 and LGALS3BP, as well as low expressions of ECI2, COL1A1, and CBR1, were associated with treatment response. In the MTX+HCQ group, high expression of RPL27A and low expression of GGT1 were associated with response. These results confirmed that the distribution of feature effect directions across individuals was consistent with those identified by LASSO. The corresponding depiction has been added to the Results and Methods sections, please refer to lines 274-276 and 554-559.

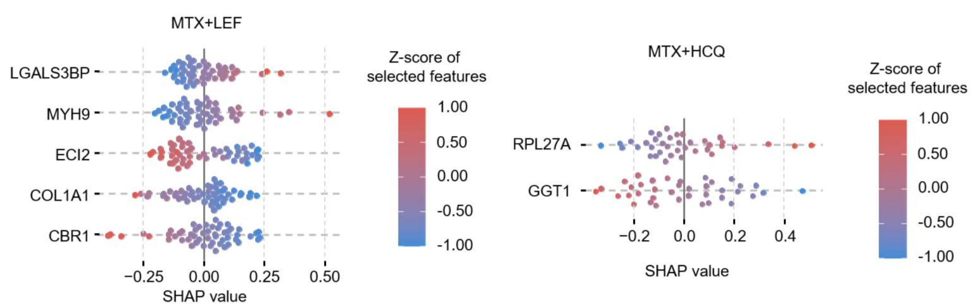

FIGURE 2. SHAP summary plot depicting feature contributions from Lasso-based models in predicting treatment response to MTX + LEF (left) and MTX + HCQ (right) (refer to Extended Data Figure 6c in the revised manuscript).